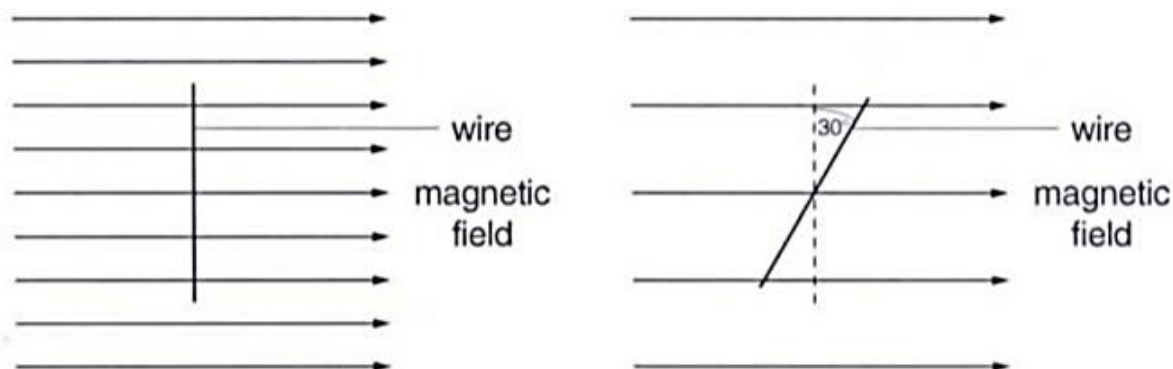


- 24** A straight, horizontal, current-carrying wire lies at right angles to a horizontal magnetic field. The field exerts a vertical force of 8.0 mN on the wire.

The wire is rotated, in its horizontal plane, through 30° as shown. The flux density of the magnetic field is halved.



What is the vertical force on the wire?

- A** 2.0 mN **B** 3.5 mN **C** 4.6 mN **D** 8.0 mN

24	B	Diagram on the left: $F = BIL\sin 90^\circ$ hence $BIL = 8.0$ mN Diagram on the right: B is halved hence $F = \frac{1}{2}BIL\sin \theta = \frac{1}{2} 8.0 \sin 60^\circ = 3.5$ mN
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25 The diagram below shows a long wire carrying a current I